Final Project Reflection Report

CS 411
Project Track 1

1) **Please list out changes in the directions of your project if the final project is different from your original proposal (based on your stage 1 proposal submission).**
   We mostly stayed true to our proposal vision. However, we did originally think of using Google Maps API to have live GGIS data that we could use to find the top k pharmacies near a user. However, we found a US pharmacies dataset that was very extensive, even to the county level. Hence, we just went ahead with querying that. However, if we had to expand the use cases outside of the US, then we would have to use APIs or more data.

2) **Discuss what you think your application achieved or failed to achieve regarding its usefulness.**
   We were able to match a user’s symptoms to a list of best rated medicines, which was our main goal. We were also able to achieve geolocating nearby pharmacies via the dataset above. One aspect that we failed at was being able to track the inventory of pharmacies nearby. However, that is something that cannot be overcome without some service that does this tracking, which all pharmacies abide by and trust, and to which we can call its API for our application.

3) **Discuss if you changed the schema or source of the data for your application.**
   We heavily used the datasets that we found online (on Kaggle) at the beginning of the semester. However, we also synthesized large data, such as user data and their associated symptoms. Only by utilizing both synthesized and the online datasets, we were able to progress with the project.

4) **Discuss what you change to your ER diagram and/or your table implementations. What are some differences between the original design and the final design? Why? What do you think is a more suitable design?**
   We did not need some features from what we had on the ER diagram. For example, all the manufacturer data is not used, and our actual code does not use that. Apart from that, we did not really stray from our ER diagram much, even in our final implementation. We believe that what we have in code is an accurate representation of our suitable design proposed in the ER diagram.

5) **Discuss what functionalities you added or removed. Why?**
Beyond the live geolocation services we had planned above, we also were thinking of reporting things such as 'Time it takes to drive to pharmacy x', by using average driving data and zip codes to figure out these metrics. However, this was too far from our original proposal goals, and we realized that we were straying too much from the actual vision of the product that we wanted to build. Thus, we stuck to just finding pharmacies, without worrying about getting to them or aspects such as the rating of the physical pharmacy itself.

6) **Explain how you think your advanced database programs complement your application.**
Advanced database programs help us do everything from querying the top rated medicines for a given combination of the user symptoms, to finding out which pharmacies are the closest with just datasets. Our app also benefits from advanced database programs because we leverage profiles and login information to store user data in our backend tables. It is interesting that we do not need another service to do this. Finally, we also have the ability to search, filter and even update through massive amounts of data to meet a specific set of constraints, and very quickly as well. Our app would not work with huge amounts of US-centric location data and medicine data, and the fact that we can generate curated lists of medicines and where to find them is powered through these advanced database programs.

7) **Each team member should describe one technical challenge that the team encountered. This should be sufficiently detailed such that another future team could use this as helpful advice if they were to start a similar project or where to maintain your project.**
Vihaan: Make sure to document how to run each part of your project as you build it up, and how that process might change depending on your device platform, so that it is not forgotten each time. For example, I encountered some Mac-specific problems that I overcame by leveraging the curl command. The final project however, should not have any device-specific problems.
Tuan: Make sure to verify whether your datasets are valid for the project and whether they are extensive/diverse enough for your needs. We ended up having to synthesize a lot of data to fill in any gaps we encountered.
Maahum: It is important to check and test extensively at each stage. For example, you might have met all the requirements for a stage, but that does not mean you have designed well enough to handle the requirements of the next stage. Read ahead and take future checkpoints into consideration while designing your project.

Ryan: Before writing any code, it is important to define what exactly you want done. For example, we wanted to write a stored procedure to get the closest pharmacies to a user address. But what is the best way to do that? We could compare differences in zip codes, or choose a nearby city with the most viable pharmacies. It is important to know what you want, and the actual code will come to you easily, using just what is taught in class.

8) **Are there other things that changed comparing the final application with the original proposal?**
Our UI changed a lot from our original vision. We were going to have a dashboard that had images of medicines and submenus that a user could use to view their past searches and purchases. We were also going to have a window for a map that showed their address with a red dot and the pharmacies they could go to with green or yellow symbols. We also planned to add features such as sickness tracking, where every time a user queried their symptoms, the app would store that on a per-month basis, so that the user could have information such as whether seasonal changes strongly affect them. However, these features were more “nice-to-have”s than what we needed within the scope of the project, and so we decided to put our efforts towards the backend and the database programs instead.

9) **Describe future work that you think, other than the interface, that the application can improve on**
For future work, on the database/backend side, we would want to add more personalization per user, where each user has their own metrics they can view, such as average time between searches (which could roughly translate to frequency of getting sick), or expenses per year on medicine. We could also add database-wide metrics, such as most frequent pharmacies picked, which could have interesting conclusions on where most users are. Hence, if we were to scale this, we could focus our server loads in those regions, while still ensuring that all users have a good baseline connectivity (as it is a medical aid platform we are trying to build). Overall, we think there is so much more we can do, both for users, and to improve the app itself, using the data that we would have (within legal/moral constraints). We can also track metrics such as “Manufacturer with the most diverse product line” or “Most revenue per manufacturer per region”, by just storing which medicines were bought, without having it associated with which user bought them. One last example is store hours, where we can simply let a user view whether the address they are headed to is open or not.

10) **Describe the final division of labor and how well you managed teamwork.**

We had a pretty even division of labor throughout the project. Vihaan handled dataset creation, table creation on the cloud studio and creating the baseline queries to access data. He also created the functionality to find the best rated medicines for a given set of symptoms. Tuan improved upon these to create some advanced stored procedures that we could use to find the top k pharmacies based on the user's address. Ryan was originally going to use the Google Maps (or another GGIS) API to geolocate, but as we transitioned to using a large dataset instead, he helped integrate that. Additionally, he designed the sequence of states of how the app should run, and also built the frontend with Tuan. Maahum designed and implemented the transactions logic and formulated medicine scoring based on reviews and prices. She also improved iterations of our stored procedures and reviewed UI design. Overall, we handled teamwork pretty well, with weekly check ins and both in-person and online work sessions.